



Name: _____

Teacher: _____

Class: _____

FORT STREET HIGH SCHOOL

2010

**HIGHER SCHOOL CERTIFICATE COURSE
ASSESSMENT TASK 1**

Mathematics Extension 1

TIME ALLOWED: 45 MINUTES

Outcomes Assessed	Questions	Marks
Applies appropriate techniques to solve problems involving parametric representations	1,2	/19
Applies appropriate techniques from the study of calculus to solve problems	3	/9

Question	1	2	3	Total	%
Marks	/5	/14	/9	/28	

Directions to candidates:

- Attempt all questions
- The marks allocated for each question are indicated
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board – approved calculators may be used
- Each new question is to be started in a new booklet

Question 1 (5 marks)

- a) Write the Cartesian equation which is described by the parametric equations
 $x = t + 1, y = \frac{1}{t}$ 2
- b) i) Find the equation of the chord of contact of the tangents from the point
 $T(3, -2)$ to the parabola $x^2 = 8y$ 1
- ii) Show that this chord of contact passes through the focus of the parabola 2

Question 2 (14 marks)

- a) $P(2ap, ap^2)$, $Q(2aq, aq^2)$, $R(2ar, ar^2)$ are points on $x^2 = 4ay$
i) If the tangent at P is parallel to the chord QR , find the gradient at P and hence
show that $q + r = 2p$ 3
- ii) M lies on QR such that $RM = MQ$ prove that PM is parallel to the axis of the
parabola. 3
- b) i) Show that the equation of the line from the focus S of the parabola $x^2 = 4ay$
and perpendicular to the normal through the point P $(2ap, ap^2)$ on the parabola is
 $px - y + a = 0$. 3
- ii) If these lines meet at N , show that the locus of N as P moves on the parabola
 $x^2 = 4ay$ is also a parabola. 4
- iii) State the vertex and focus of the locus. 1

Question 3 (9 marks)

- a) i) Show that there is a root for $f(x) = x^3 - 3x - 4$ in the domain $2 \leq x \leq 3$ 2
- ii) Use two applications for Halving the Interval method to determine an
approximation for the root correct to 2 decimal places 3
- iii) Use one application of Newton's method to determine an approximation for the
root correct to 2 decimal places 1
- b) Take $x = 1.3$ and use two applications of Newton's method to find the cube root of
2. Give answer to 2 decimal places. 3

Q1 a) $x = t+1$, $y = \frac{1}{t}$
 $t = x-1$

$\therefore y = \frac{1}{x-1}$

b) i)

Chord of contact is $xx_1 = 2a(y+y_1)$

$x^2 = 8y \therefore a = 2$ $x_1 = 3$, $y_1 = -2$

$\therefore 3x = 4(y-2)$

$3x = 4y - 8$

$3x - 4y + 8 = 0$

ii) Focus is at $(0, 2)$

i.e. $0 - 8 + 8 = 0$ True

so $3x - 4y + 8 = 0$ passes through the focus.

✓ 1 for co-ord. of focus
 1 for subst. into eq.

Q2 a)

i) $\frac{x^2}{4a} = y$

$y' = \frac{x}{2a}$ and $(2ap, ap^2)$

$m = \frac{2ap}{2a}$
 $= p$

1 for correctly obtained gradient by calculus.

Since tangent is parallel to QR

$p = m \text{ of QR}$

$m = \frac{aq^2 - ar^2}{2aq - 2ar}$

$= \frac{a(q-r)(q+r)}{2a(q-r)}$

$= \frac{q+r}{2}$

$\therefore p = \frac{q+r}{2}$

$2p = q+r$

✓

✓

✓

M is midpoint of QR

$$\therefore M \text{ is } \left(\frac{2aq+2ar}{2}, \frac{aq^2+ar^2}{2} \right)$$

$$\left(\frac{2a(q+r)}{2}, \frac{a(q^2+r^2)}{2} \right) \checkmark$$

$$(2ap, \frac{a(q^2+r^2)}{2}), q+r=2p$$

$$PM \text{ has } m = \frac{\frac{a(q^2+r^2)}{2} - \frac{2ap^2}{2}}{2ap - 2ap} \checkmark$$

i.e. m is undefined and

PM is a vertical line which is parallel to $x=0$, the axis of the parabola. $\checkmark$

b) i) $x^2=4ay$
 $y' = \frac{2x}{4a}$

and at P m of normal $= -\frac{1}{p}$ $\checkmark$

Line $\perp$ to normal has $m=p$

Eq is $y-y_1 = m(x-x_1)$ $\checkmark$

$$y-a = p(x-0) \text{ at focus}$$

$$y-a-px=0$$

$$px-y+a=0 \quad \textcircled{1} \checkmark$$

ii) Eq of normal is

$$y-ap^2 = -\frac{1}{p}(x-2ap)$$

$$py-ap^3 = -x+2ap$$

$$x+py = 2ap+ap^3 \quad \textcircled{2}$$

at N $px+p^2y = 2ap^2+ap^4$ from $\textcircled{2}$

$$y-a+p^2y = 2ap^2+ap^4 \text{ from } \textcircled{1}$$

1 for midpoint

Most students understood concept but gave very poor explanations

1 for parallel lines using gradient information

Could also say M and P have same x value $\therefore$ lie on $x=2ap$ which is $\parallel$ to $x=0$.

1 for gradient

1 for eq. at focus

1 for equation correctly obtained

$\checkmark$ 1 for equation of normal.

Algebraic errors cost marks here

$$y(p^2) = 1$$

$$= a(p^2 + 1)^2$$

$$y = a(p^2 + 1)$$

$$\therefore x = \frac{a(p^2 + 1) - a}{p}$$

$$= \frac{a(p^2 + 1 - 1)}{p}$$

$$\frac{x}{a} = p \text{ i.e. } N(ap, a(p^2 + 1)^2)$$

$$\text{so } y = a \left[\left(\frac{x}{a} \right)^2 + 1 \right]$$

$$y - a = \frac{x^2}{a}$$

$$x^2 = ay - a^2$$

$$= a(y - a)$$

$\therefore$ locus is a parabola with vertex at $(0, a)$ and focal length $\frac{a}{4}$

$\therefore$ Focus $(0, \frac{5a}{4})$

Q3

a) i) $f(x) = x^3 - 3x - 4$

$$f(2) = 8 - 6 - 4 < 0$$

$$f(3) = 27 - 9 - 4 > 0$$

$\therefore$ a root lies in $2 \leq x \leq 3$.
Since $f(x)$ is continuous and there is a change of sign.

ii) $f(2.5) = (2.5)^3 - 3(2.5) - 4$
 $= 4.125$

$f(2.5) > 0$ and $f(2) < 0$
so root lies between 2 and 2.5

$$f(2.25) = (2.25)^3 - 3(2.25) - 4$$

$$= 0.690625$$

$f(2.25) > 0$ and $f(2) < 0$
 $\therefore$ so root lies between 2 and 2.25

$$\therefore x = 2.25$$

Many did not factorise

Many students failed to realize that here $a = 4 \times \text{focal length}$
 $\therefore \text{focal length} = \frac{a}{4}$

1 for both correct

Students must learn to fully execute two applications rather than find $f(x)$ values for several values of x & then come up with an answer.

Final approxn after two applications is $x = 2.25$ as $f(2.25)$ is closer to zero than $f(2)$.
DO NOT HAVE A THIRD TIME.

1 for statement of the root approx.

$$f'(x) = 3x^2 - 3, \quad f'(2.13) = 10.6107$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= 2.13 + \frac{0.726403}{10.6107}$$

$$\approx 2.19845948$$

$$x = 2.20 \text{ (to 2 d.p.)}$$

1 for
use of
Newton's
method
correctly.

Some chose
a value right
outside the
range $2 < x < 2.25$
established in
(ii).

b) $x^3 = 2$
 $x^3 - 2 = f(x)$
 $3x^2 = f'(x)$

$$x = 1.3$$

$$x_2 = 1.3 - \frac{1.3^3 - 2}{3(1.3)^2}$$

$$\approx 1.261143984$$

$$x_3 = 1.261143984 - \frac{(1.261143984^3 - 2)}{3(1.261143984)^2}$$

$$\approx 1.259922235$$

$$x = 1.26 \text{ (to 2 d.p.)}$$

1 establishing
 $f(x)$.

Some students
used
 $x = \sqrt[3]{2}$

$$\therefore f(x) = x - \sqrt[3]{2}$$

$$+ f'(x) = 1$$

which simplified
process.
(gave their full
marks!)

Must use full digit
display when re-using
an answer